

RESEARCH PITCH | EMPIRICAL STUDY

Model Instability Under Minority Data Scarcity

An Empirical Study of Model Capacity in Imbalanced Classification

A structured research roadmap covering datasets, models, metrics,
existing baselines, and the proposed **AMBA** framework

Status: Research Proposal / Pitch

Date: 2026

Abstract

Class imbalance—the condition where one class vastly outnumbers another—is one of the most common and consequential obstacles in applied machine learning. Real-world high-stakes tasks, including fraud detection, rare disease diagnosis, and network intrusion identification, suffer from severe imbalance. Existing work predominantly reports average performance improvements under standard resampling or cost-sensitive methods, without closely examining the *stability* of model predictions across experimental runs, data splits, and varying imbalance ratios.

This document pitches a structured empirical study with three goals. First, we benchmark seven commonly used classifiers across four carefully chosen datasets spanning a wide range of imbalance intensities, using both performance and stability metrics. Second, we apply a representative set of existing imbalance-handling techniques as baselines. Third, we introduce **AMBA** (**A**daptive **M**inority **B**oundary **A**ugmentation)—a novel framework that combines imbalance-ratio-aware routing, uncertainty-guided synthetic oversampling with a scarcity-proportional budget, and stability-weighted ensemble aggregation. AMBA is designed from the ground up to address prediction instability that existing techniques overlook, while explicitly adapting its strategy to the imbalance severity and separability of the data.

No experimental results are reported in this document; the goal is to establish the research question, motivate the experimental choices, and describe the proposed methodology with sufficient mathematical and algorithmic precision to serve as a working blueprint.

0 Contents

1	Introduction and Motivation	3
1.1	Why Instability Is the Overlooked Problem	3
1.2	Scope of This Study	3
2	Problem Formulation	3
2.1	Formal Setup	3
2.2	The Learning Objective Under Imbalance	4
2.3	Defining Stability	4
3	Candidate Datasets	4
3.1	Dataset Descriptions	4
4	Candidate Models	5
5	Evaluation Metrics	5
5.1	Why Accuracy Alone Fails	5
5.2	Confusion Matrix Foundations	5
5.3	Performance Metrics	6
5.4	Stability Metrics	6
6	Existing Imbalance-Handling Techniques (Baselines)	6
6.1	Data-Level Techniques	7
6.2	Algorithm-Level Techniques	7
6.3	Ensemble-Level Techniques	7
7	Proposed Technique: AMBA	7
7.1	Motivation	8
7.2	Component 0: Adaptive Pre-Flight Routing	8
7.3	Stage 1: Adaptive Boundary Detection	8
7.4	Stage 2: Budget-Capped Directionally Guided Synthetic Oversampling	9
7.5	Stage 3: Stability-Weighted Ensemble	10

7.6	Complete Pipeline Overview	11
7.7	AMBA Pseudocode	12
7.8	Hyperparameters and Their Roles	13
7.9	What AMBA Does That Existing Methods Do Not	13
8	Experimental Roadmap	13
8.1	Experimental Design	13
8.2	Phase 1: Data Preparation	14
8.3	Phase 2: Baseline Evaluation	14
8.4	Phase 3: Existing Technique Baselines	14
8.5	Phase 4: AMBA Evaluation	14
8.6	Phase 5: Analysis	14
9	Expected Contributions	15
10	Conclusion	15

1 Introduction and Motivation

In many real-world classification tasks, the distribution of class labels is far from uniform. A dataset used to detect fraudulent credit card transactions may contain 99.8 % legitimate records and only 0.2 % fraud. A medical screening dataset may contain 95 % healthy patients and 5 % at-risk individuals. This asymmetry—called *class imbalance*—creates a fundamental challenge: a naive classifier that always predicts the majority class can achieve very high accuracy while being completely useless in practice.

1.1 Why Instability Is the Overlooked Problem

The machine learning community has responded to class imbalance with a large toolkit: oversampling, undersampling, cost-sensitive learning, and ensemble methods. Most benchmark studies report that these techniques improve the mean F1-score or AUC-ROC over a baseline. However, three important questions are rarely asked:

1. Does the performance improvement come with *consistent* results, or does the improved mean hide high variance across folds and runs?
2. Do different classifier architectures behave differently in terms of stability when minority data is scarce?
3. Can we deliberately engineer a method that optimises *both* the level and the stability of minority-class performance, while adapting its strategy to the regime of imbalance it encounters?

Central Claim. A technique that achieves a slightly lower mean F1-score but with half the variance is often more operationally useful than a technique with a higher mean but unpredictable individual-run behaviour. We argue that stability is a first-class evaluation criterion, not an afterthought—and that an effective imbalance-handling framework must adapt its augmentation strategy to the structure of the data it receives.

1.2 Scope of This Study

We restrict ourselves to tabular, binary classification problems. This scope is chosen deliberately: tabular data remains the dominant form in industry, binary classification is the clearest setting for studying minority-class behaviour, and it avoids confounds from domain-specific representation learning (e.g., image or text encoders).

2 Problem Formulation

2.1 Formal Setup

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ be a labelled dataset where $\mathbf{x}_i \in \mathbb{R}^d$ is a feature vector and $y_i \in \{0, 1\}$ is the class label. We denote the majority class as $y = 0$ and the minority class as $y = 1$. Define:

$$D^+ = \{(\mathbf{x}_i, y_i) \mid y_i = 1\}, \quad D^- = \{(\mathbf{x}_i, y_i) \mid y_i = 0\}.$$

The *Imbalance Ratio* is:

$$\text{IR} = \frac{|D^-|}{|D^+|}.$$

A dataset is considered *mildly* imbalanced if $\text{IR} < 5$, *moderately* imbalanced if $\text{IR} \in [5, 50]$, and *severely* or *extremely* imbalanced if $\text{IR} > 100$.

2.2 The Learning Objective Under Imbalance

A standard classifier minimises the empirical risk:

$$\hat{f} = \arg \min_{f \in \mathcal{F}} \frac{1}{N} \sum_{i=1}^N \ell(f(\mathbf{x}_i), y_i),$$

where ℓ is a loss function. Under standard 0-1 loss and high imbalance, this optimisation is biased toward the majority class because majority errors dominate the sum. As a result, $\hat{f}$ tends to underperform on D^+ .

2.3 Defining Stability

Let $\mathcal{S} = \{S_1, S_2, \dots, S_K\}$ be a set of cross-validation folds. Let m_k be a performance metric (e.g., recall or F1-score on the minority class) computed on fold k . We define the fold mean and standard deviation:

$$\bar{m} = \frac{1}{K} \sum_{k=1}^K m_k, \quad \sigma_m = \sqrt{\frac{1}{K-1} \sum_{k=1}^K (m_k - \bar{m})^2}.$$

The *Coefficient of Variation* (CV) is:

$$\text{CV}(m) = \frac{\sigma_m}{\bar{m}}.$$

A lower CV indicates greater stability. Our study will report both $\bar{m}$ and $\text{CV}(m)$ for all model–technique combinations.

3 Candidate Datasets

We select four publicly available binary classification datasets spanning a wide spectrum of imbalance ratios, sample sizes, and application domains. This diversity is intentional: a good technique should generalise across settings.

Table 1: Summary of candidate datasets ordered by increasing imbalance ratio.

Dataset	N	Features	IR	Category	Domain
Pima Indians Diabetes	768	8	≈ 1.9	Mild	Biomedical
Sick (Thyroid Disease, OpenML)	3,772	29	≈ 15	Moderate	Medical
Mammography (OpenML)	11,183	6	≈ 42	High	Medical Imaging
Credit Card Fraud (ULB)	284,807	30	≈ 578	Extreme	Finance

3.1 Dataset Descriptions

1. Pima Indians Diabetes (IR ≈ 1.9). Sourced from the UCI Machine Learning Repository, this dataset records eight clinical measurements from Pima Indian women alongside a binary diabetes indicator. With mild imbalance it serves as the near-baseline condition, showing model behaviour before minority scarcity becomes severe.

2. Sick Dataset (IR ≈ 15). From OpenML, Clinical thyroid screening dataset with multiple laboratory and diagnostic features, reduced to a binary classification problem. The moderate imbalance and relatively large sample size make it well-suited for studying model stability and performance under intermediate skew, bridging the gap between near-balanced and highly imbalanced regimes.

3. Mammography Dataset ($IR \approx 42$). Available on OpenML (dataset ID 310), this records pixel features from mammography scans with malignancy as the minority class. The high IR and small minority count (≈ 260 samples) make it a realistic test of techniques under genuine data scarcity.

4. Credit Card Fraud Detection ($IR \approx 578$). Released by the ULB Machine Learning Group, this contains transaction-level features from European cardholders with 492 fraud cases in 284,807 total transactions. The extreme imbalance and large scale make it the hardest dataset in our suite.

4 Candidate Models

We select seven classifiers spanning a wide range of inductive biases, capacities, and decision-boundary geometries. This diversity is necessary to study how model type interacts with imbalance and with different remediation techniques.

Table 2: Summary of the seven candidate models.

ID	Model	Rationale for Inclusion
M1	Logistic Regression (LR)	Linear boundary; well-calibrated probabilities; directly sensitive to class priors
M2	Support Vector Machine (SVM, RBF)	Margin-based; implicit regularisation; known to struggle with severe imbalance
M3	Decision Tree (DT)	Axis-aligned boundaries; prone to majority-dominated splits
M4	Random Forest (RF)	Ensemble of trees; partial robustness through bagging
M5	Gradient Boosted Trees (XGBoost)	Sequential boosting; supports controllable class weights
M6	Multi-layer Perceptron (MLP)	Flexible non-linear model; highly sensitive to minority sample count
M7	K-Nearest Neighbours (KNN)	Non-parametric; directly affected by neighbourhood density imbalance

5 Evaluation Metrics

5.1 Why Accuracy Alone Fails

On a dataset with $IR = 100:1$, a classifier that always predicts the majority class achieves 99% accuracy. This makes standard accuracy an unreliable metric under imbalance; we require metrics sensitive to minority-class performance.

5.2 Confusion Matrix Foundations

For binary classification, the confusion matrix defines four counts: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). Under imbalance, FN (missed minority samples) is typically the most costly error.

5.3 Performance Metrics

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (1)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (2)$$

$$\text{F1-score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \text{ TP}}{2 \text{ TP} + \text{FP} + \text{FN}} \quad (3)$$

$$\text{AUC-ROC} = \int_0^1 \text{TPR}(t) \, d\text{FPR}(t) \quad (4)$$

$$\text{G-Mean} = \sqrt{\text{Sensitivity} \times \text{Specificity}} = \sqrt{\frac{\text{TP}}{\text{TP} + \text{FN}} \cdot \frac{\text{TN}}{\text{TN} + \text{FP}}} \quad (5)$$

$$\text{MCC} = \frac{\text{TP} \cdot \text{TN} - \text{FP} \cdot \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}} \quad (6)$$

Note on MCC. Matthews Correlation Coefficient considers all four confusion matrix cells and is invariant to class prevalence. Its range is $[-1, +1]$, where $+1$ is perfect, 0 is no better than random, and -1 is perfectly wrong.

5.4 Stability Metrics

In addition to point-estimate performance metrics, we report the following stability quantities computed over K -fold cross-validation:

$$\text{CV}(\text{F1}) = \frac{\sigma_{\text{F1}}}{\bar{F}_1} \quad (\text{Coefficient of Variation of F1}) \quad (7)$$

$$\text{CV}(\text{Recall}) = \frac{\sigma_{\text{Recall}}}{\bar{\text{Recall}}} \quad (8)$$

$$\text{IQR}(m) = Q_{75}(m) - Q_{25}(m) \quad (\text{Interquartile Range of metric } m) \quad (9)$$

Table 3: Complete metric suite. The left block measures performance level; the right block measures consistency across folds.

Performance Metrics			Stability Metrics		
Metric	Summary	Goal	Metric	Measures	Goal
Precision	$\text{TP}/(\text{TP}+\text{FP})$	$\uparrow 1$	CV(F1)	F1 consistency	$\downarrow 0$
Recall	$\text{TP}/(\text{TP}+\text{FN})$	$\uparrow 1$	CV(Recall)	Recall consistency	$\downarrow 0$
F1-score	Harmonic mean (P, R)	$\uparrow 1$	$\sigma(\text{MCC})$	MCC spread	$\downarrow 0$
AUC-ROC	Area under ROC	$\uparrow 1$	IQR(F1)	Fold robustness	$\downarrow 0$
G-Mean	$\sqrt{\text{Sens} \times \text{Spec}}$	$\uparrow 1$			
MCC	Balanced correlation	$\uparrow 1$			

6 Existing Imbalance-Handling Techniques (Baselines)

We organise existing techniques into three families. Each will be applied to each of the seven models across all four datasets, yielding a comprehensive baseline surface for comparison.

6.1 Data-Level Techniques

Random Oversampling. Minority samples are duplicated uniformly at random until a target ratio is reached.

Random Undersampling. Majority samples are removed uniformly at random to reduce the imbalance, at the risk of discarding useful majority-class information.

SMOTE. For each $\mathbf{x}_i \in D^+$, SMOTE finds its k nearest minority neighbours and generates a synthetic sample by linear interpolation:

$$\mathbf{x}_{\text{syn}} = \mathbf{x}_i + \lambda (\mathbf{x}_{\text{nn}} - \mathbf{x}_i), \quad \lambda \sim \mathcal{U}[0, 1].$$

ADASYN. Generates more samples for harder minority instances. The difficulty of $\mathbf{x}_i$ is proportional to the number of majority neighbours in its k -neighbourhood:

$$r_i = \frac{\Delta_i}{k}, \quad \hat{r}_i = \frac{r_i}{\sum_j r_j}.$$

Tomek Links. A cross-class pair $(\mathbf{x}_i, \mathbf{x}_j)$ is a Tomek Link if each is the other's nearest neighbour. Removing the majority-class member from each link cleans the decision boundary region.

6.2 Algorithm-Level Techniques

Class Weighting. The loss is scaled to penalise minority errors more heavily:

$$\ell_w(\hat{y}, y) = w_y \cdot \ell(\hat{y}, y), \quad w_1 = \frac{N}{2|D^+|}, \quad w_0 = \frac{N}{2|D^-|}.$$

Threshold Moving. Rather than using the default 0.5 decision threshold, τ^* is tuned on a validation set to maximise a target metric:

$$\hat{y} = \mathbf{1}[P(y = 1 \mid \mathbf{x}) \geq \tau^*], \quad \tau^* = \arg \max_{\tau} \text{G-Mean}(\tau).$$

6.3 Ensemble-Level Techniques

Balanced Bagging. Multiple classifiers are trained on bootstrapped subsets, each balanced by undersampling the majority class to match the minority count.

EasyEnsemble. An ensemble of AdaBoost classifiers is trained on different random majority-class undersamples, with final predictions by majority vote.

7 Proposed Technique: AMBA

AMBA: Adaptive Minority Boundary Augmentation A four-component framework that (1) routes the pipeline according to the imbalance ratio and baseline class separability; (2) uses an imbalance-adaptive initial classifier to locate minority samples near the decision boundary; (3) generates synthetic samples in this boundary region using gradient-directed interpolation within a scarcity-proportional budget; and (4) aggregates an ensemble of classifiers weighted by their cross-validated stability rather than their mean performance alone.

7.1 Motivation

Existing oversampling and ensemble methods share three key limitations:

1. **Geometrically blind augmentation.** SMOTE generates synthetic samples uniformly in the minority neighbourhood without regard for where the decision boundary lies, potentially reinforcing already-covered regions while leaving the boundary zone sparse.
2. **Ignoring instability as a design criterion.** No existing technique explicitly minimises prediction variance as part of its objective.
3. **One-size-fits-all strategy.** Existing techniques apply the same augmentation logic regardless of the imbalance ratio or the intrinsic separability of the classes. At mild imbalance the minority class is already well-represented; heavy augmentation introduces noise. At extreme imbalance, unconstrained synthesis risks mode collapse. When classes are already well-separated, augmentation adds variance, not signal.

AMBA addresses all three limitations through an adaptive routing system and a scarcity-aware synthesis budget.

7.2 Component 0: Adaptive Pre-Flight Routing

Before entering the main pipeline, AMBA applies two routing checks.

IR-Adaptive Bypass (handles mild imbalance)

$$\text{If } \text{IR} < \tau_{\text{IR}} : \quad \hat{y} = 1$$

$$b|\mathbf{x}) \geq \tau^*$$

$\tau^* = \arg \max_{\tau} \text{G-Mean}(\tau, D_{\text{val}})$, where f is a standard cross-validated classifier with no augmentation. At mild imbalance ($\text{IR} < \tau_{\text{IR}}$, default: 5) the minority class is naturally well-represented across folds. The overhead of boundary detection, guided synthesis, and ensemble construction introduces variance without benefit. Threshold Moving is the appropriate lightweight alternative here.

AUC-Gated Augmentation Bypass (handles near-perfectly separated classes)

After a pilot 3-fold cross-validation on the raw data, compute the mean baseline AUC. If $\text{AUC}_{\text{baseline}} \geq \theta_{\text{AUC}}$ (default: 0.97), skip Stages 1 and 2 (no synthetic oversampling) and execute Stage 3 only: train the stability-weighted ensemble on the raw class-weighted data. This retains the stability benefit of ensemble weighting while avoiding the noise from augmenting an already-solved classification problem.

When the baseline AUC exceeds 0.97, the model has effectively learned the decision boundary. Boundary-directed synthesis in Stage 2 would be solving a problem that does not exist.

7.3 Stage 1: Adaptive Boundary Detection

Train an initial classifier f_0 on the raw imbalanced data $\mathcal{D}$ using an IR-conditioned selector:

$$f_0 = \begin{cases} \text{Logistic Regression} & \text{if } \text{IR} < \tau_{f_0} \\ \text{Random Forest (shallow, max_depth = 4)} & \text{if } \text{IR} \geq \tau_{f_0} \end{cases}$$

with τ_{f_0} defaulting to 20. At high IR, the minority samples form a sparse, possibly non-convex cluster that a linear boundary cannot fit. A shallow Random Forest captures non-linear boundaries without overfitting and provides well-calibrated minority-class probabilities after Platt scaling.

For each $\mathbf{x}_i \in D^+$, compute its boundary proximity score:

$$b_i = 1 - 2 \left| P_{f_0}(y = 1 \mid \mathbf{x}_i) - 0.5 \right| \in [0, 1].$$

$b_i \approx 1$ means $\mathbf{x}_i$ is near the boundary (the model is uncertain); $b_i \approx 0$ means it lies well inside the minority region.

Partition D^+ at threshold τ_b :

$$D_{\text{bnd}}^+ = \{\mathbf{x}_i \in D^+ : b_i \geq \tau_b\}, \quad D_{\text{safe}}^+ = \{\mathbf{x}_i \in D^+ : b_i < \tau_b\}.$$

Gradient estimation. When f_0 is Logistic Regression, the closed-form gradient is available:

$$\mathbf{g}_i = -\nabla_{\mathbf{x}} P_{f_0}(y = 1 \mid \mathbf{x}) \big|_{\mathbf{x}=\mathbf{x}_i}.$$

When f_0 is a Random Forest, we use a finite-difference directional estimate:

$$\hat{g}_{i,j} \approx \frac{P_{f_0}(y = 1 \mid \mathbf{x}_i + \epsilon \mathbf{e}_j) - P_{f_0}(y = 1 \mid \mathbf{x}_i)}{\epsilon}, \quad \epsilon = 10^{-3} \sigma_j,$$

for each feature dimension j , where σ_j is the feature standard deviation. The full gradient vector is normalised: $\hat{\mathbf{g}}_i = \mathbf{g}_i / \|\mathbf{g}_i\|$.

7.4 Stage 2: Budget-Capped Directionally Guided Synthetic Oversampling

Scarcity-proportional budget cap

The total synthetic sample budget is constrained to prevent mode collapse at extreme imbalance:

$$n_{\text{syn}}^{\text{max}} = \min(n_{\text{syn}}, \lfloor \gamma \cdot |D^+| \rfloor), \quad \gamma \geq 1 \text{ (default: 3.0)}.$$

This ensures the synthetic set cannot be more than γ times the real minority set size. At extreme IR with small $|D^+|$, unconstrained synthesis creates a bloated but low-diversity minority cloud that collapses ensemble diversity and sacrifices precision. The budget is allocated as:

$$n_{\text{syn}}^{\text{bnd}} = \lfloor \rho \cdot n_{\text{syn}}^{\text{max}} \rfloor, \quad n_{\text{syn}}^{\text{safe}} = n_{\text{syn}}^{\text{max}} - n_{\text{syn}}^{\text{bnd}}, \quad \rho \in (0.5, 1).$$

Boundary-directed synthesis (D_{bnd}^+ samples)

The gradient of f_0 's confidence is incorporated to direct interpolation toward the decision boundary:

$$\mathbf{x}_{\text{syn}} = \mathbf{x}_i + \lambda_1 (\mathbf{x}_{\text{nn}} - \mathbf{x}_i) + \alpha \lambda_2 \hat{\mathbf{g}}_i,$$

where $\hat{\mathbf{g}}_i = \mathbf{g}_i / \|\mathbf{g}_i\|$, $\lambda_1, \lambda_2 \sim \mathcal{U}[0, 1]$ independently, and $\alpha \geq 0$ controls the boundary-direction contribution.

Diversity filter

After generating each candidate $\mathbf{x}_{\text{syn}}$, reject it if:

$$\min_{j \in D_{\text{bnd}}^+} \|\mathbf{x}_{\text{syn}} - \mathbf{x}_j\| < \delta,$$

where δ is a feature-space proximity threshold (default: $0.01 \cdot \bar{\sigma}_{\mathcal{D}}$, the mean feature standard deviation). This prevents duplicate or near-duplicate synthetic samples, which would otherwise create correlated ensemble members.

Safe-region synthesis (D_{safe}^+ samples)

Standard SMOTE is applied at a lower oversampling rate, since these regions are already well-covered.

The augmented dataset is: $D_{\text{aug}} = \mathcal{D} \cup \mathcal{S}_{\text{bnd}} \cup \mathcal{S}_{\text{safe}}$.

7.5 Stage 3: Stability-Weighted Ensemble

Train K base classifiers $\{f_1, \dots, f_K\}$ on different augmented bootstrap samples. For each f_k , compute minority recall on each of V validation folds:

$$r_k^{(v)} = \frac{\text{TP}_k^{(v)}}{\text{TP}_k^{(v)} + \text{FN}_k^{(v)}}, \quad v = 1, \dots, V.$$

Compute the recall instability of classifier k :

$$\sigma_k = \sqrt{\frac{1}{V-1} \sum_{v=1}^V (r_k^{(v)} - \bar{r}_k)^2}.$$

Assign inverse-instability weights and normalise:

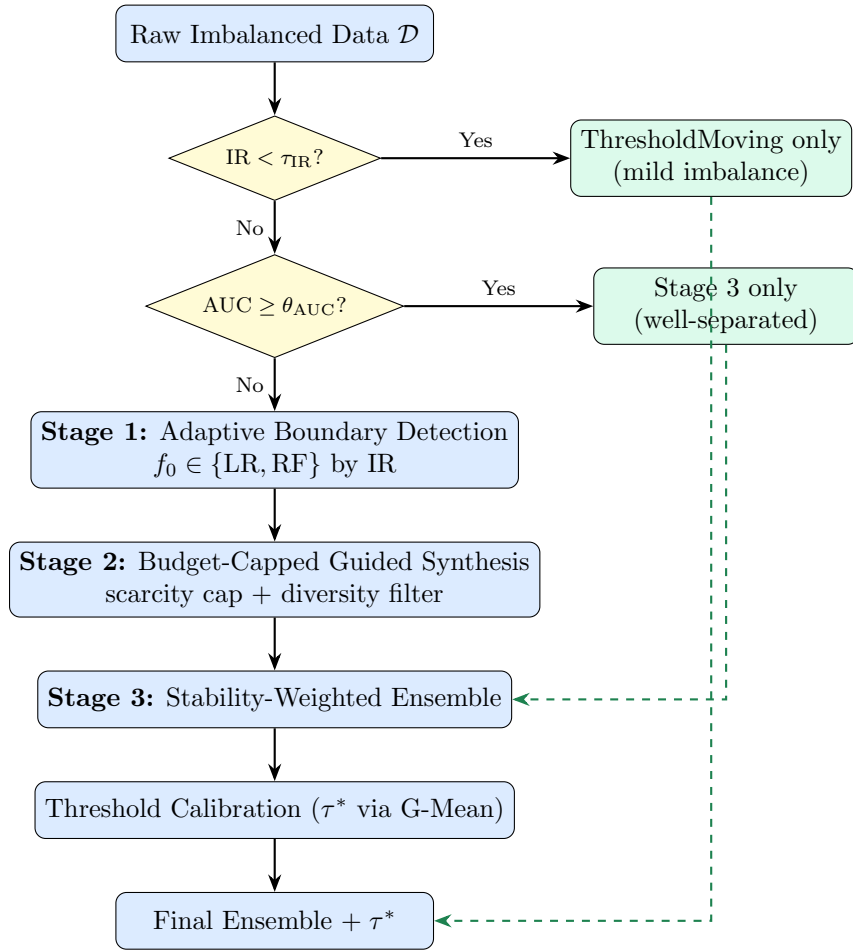
$$\tilde{w}_k = \frac{1}{1 + \sigma_k}, \quad w_k = \frac{\tilde{w}_k}{\sum_{j=1}^K \tilde{w}_j}.$$

The final prediction is a stability-weighted soft vote:

$$P_{\text{AMBA}}(y = 1 \mid \mathbf{x}) = \sum_{k=1}^K w_k \cdot P_{f_k}(y = 1 \mid \mathbf{x}), \quad \hat{y} = \mathbf{1}[P_{\text{AMBA}} \geq \tau^*],$$

where τ^* is tuned on a held-out validation set to maximise G-Mean.

7.6 Complete Pipeline Overview



7.7 AMBA Pseudocode

Algorithm 1 AMBA: Adaptive Minority Boundary Augmentation

Require: Dataset $\mathcal{D}$; hyperparameters $\tau_{\text{IR}}, \tau_{f_0}, \tau_b, \alpha, \rho, \gamma, \delta, K, V, \theta_{\text{AUC}}$

Ensure: Ensemble $\{(f_k, w_k)\}_{k=1}^K$; threshold τ^*

```

1: // Component 0: Adaptive Routing
2:  $\text{IR} \leftarrow |D^-|/|D^+|$ 
3: if  $\text{IR} < \tau_{\text{IR}}$  then
4:    $f \leftarrow \text{CrossValidatedClassifier}(\mathcal{D})$ 
5:    $\tau^* \leftarrow \arg \max_{\tau} \text{GMean}(P_f, \tau, D_{\text{val}})$ 
6:   return single classifier  $f$  with threshold  $\tau^*$ 
7: end if
8:  $\text{AUC} \leftarrow$  3-fold cross-validated AUC on  $\mathcal{D}$ 
9:  $\text{augment} \leftarrow (\text{AUC} < \theta_{\text{AUC}})$ 
10: if  $\text{augment}$  then
11:   // Stage 1: Adaptive Boundary Detection
12:   if  $\text{IR} \geq \tau_{f_0}$  then
13:      $f_0 \leftarrow \text{RandomForest}(\text{max\_depth} = 4, \mathcal{D})$ ; apply Platt scaling
14:      $\mathbf{g}_i \leftarrow \text{FiniteDiff}(P_{f_0}, \mathbf{x}_i, \epsilon)$  for all  $\mathbf{x}_i \in D^+$ 
15:   else
16:      $f_0 \leftarrow \text{LogisticRegression}(\mathcal{D})$ 
17:      $\mathbf{g}_i \leftarrow -\nabla_{\mathbf{x}} P_{f_0}(y = 1 \mid \mathbf{x})|_{\mathbf{x}_i}$  for all  $\mathbf{x}_i \in D^+$ 
18:   end if
19:    $b_i \leftarrow 1 - 2|P_{f_0}(y = 1 \mid \mathbf{x}_i) - 0.5|$  for all  $\mathbf{x}_i \in D^+$ 
20:    $D_{\text{bnd}}^+ \leftarrow \{\mathbf{x}_i : b_i \geq \tau_b\}$ ;  $D_{\text{safe}}^+ \leftarrow \{\mathbf{x}_i : b_i < \tau_b\}$ 
21:   // Stage 2: Budget-Capped Guided Synthesis
22:    $n_{\text{syn}} \leftarrow \min(n_{\text{syn}}, \lfloor \gamma \cdot |D^+| \rfloor)$  ▷ scarcity cap
23:    $n^{\text{bnd}} \leftarrow \lfloor \rho \cdot n_{\text{syn}} \rfloor$ ;  $n^{\text{safe}} \leftarrow n_{\text{syn}} - n^{\text{bnd}}$ 
24:    $\mathcal{S}_{\text{bnd}} \leftarrow \emptyset$ 
25:   for  $n^{\text{bnd}}$  iterations do
26:     Sample  $\mathbf{x}_i \sim D_{\text{bnd}}^+$ ; sample  $\mathbf{x}_{\text{nn}}$  from  $k\text{-NN}(\mathbf{x}_i, D^+)$ 
27:      $\lambda_1, \lambda_2 \sim \mathcal{U}[0, 1]$ 
28:      $\mathbf{x}_{\text{syn}} \leftarrow \mathbf{x}_i + \lambda_1(\mathbf{x}_{\text{nn}} - \mathbf{x}_i) + \alpha \lambda_2 \hat{\mathbf{g}}_i$ 
29:     if  $\min_j \|\mathbf{x}_{\text{syn}} - \mathbf{x}_j\| \geq \delta$  for  $\mathbf{x}_j \in D_{\text{bnd}}^+$  then ▷ diversity filter
30:        $\mathcal{S}_{\text{bnd}} \leftarrow \mathcal{S}_{\text{bnd}} \cup \{(\mathbf{x}_{\text{syn}}, 1)\}$ 
31:     end if
32:   end for
33:    $\mathcal{S}_{\text{safe}} \leftarrow \text{SMOTE}(D_{\text{safe}}^+, n^{\text{safe}})$ 
34:    $D_{\text{aug}} \leftarrow \mathcal{D} \cup \mathcal{S}_{\text{bnd}} \cup \mathcal{S}_{\text{safe}}$ 
35: else
36:    $D_{\text{aug}} \leftarrow \mathcal{D}$  ▷ AUC-bypass: class-weighted, no augmentation
37: end if
38: // Stage 3: Stability-Weighted Ensemble
39: for  $k = 1, \dots, K$  do
40:    $\mathcal{D}_k \leftarrow \text{Bootstrap}(D_{\text{aug}})$ ; train  $f_k$  on  $\mathcal{D}_k$ 
41:    $r_k^{(1)}, \dots, r_k^{(V)} \leftarrow V\text{-fold minority recall on } D_{\text{aug}}$ 
42:    $\sigma_k \leftarrow \text{std}(r_k^{(1)}, \dots, r_k^{(V)})$ ;  $\tilde{w}_k \leftarrow 1/(1 + \sigma_k)$ 
43: end for
44:  $w_k \leftarrow \tilde{w}_k / \sum_j \tilde{w}_j$  for all  $k$ 
45:  $\tau^* \leftarrow \arg \max_{\tau} \text{GMean}(\sum_k w_k P_{f_k}(y = 1 \mid \mathbf{x}), \tau, D_{\text{val}})$ 
46: return  $\{(f_k, w_k)\}_{k=1}^K, \tau^*$ 

```

7.8 Hyperparameters and Their Roles

Table 4: AMBA hyperparameters, their roles, defaults, and search ranges.

Parameter	Role	Default	Search Range
τ_b	Boundary proximity threshold	0.6	$\{0.4, 0.5, 0.6, 0.7, 0.8\}$
α	Gradient direction weight	0.3	$[0, 1]$
ρ	Boundary-region sample fraction	0.7	$[0.5, 0.9]$
K	Ensemble size	10	$\{5, 10, 20\}$
V	Stability estimation folds	5	$\{3, 5, 10\}$
k -NN	SMOTE neighbourhood size	5	$\{3, 5, 7, 10\}$
τ_{IR}	IR threshold for pre-flight bypass	5	$\{3, 5, 8, 10\}$
τ_{f_0}	IR threshold for f_0 selector (LR $\rightarrow$ RF)	20	$\{10, 20, 30, 50\}$
ϵ	Finite-difference step size (RF gradients)	$10^{-3}\bar{\sigma}$	—
γ	Scarcity cap multiplier ($n_{\text{syn}}^{\max} = \gamma D^+ $)	3.0	$\{1.5, 2, 3, 5\}$
δ	Min diversity distance	$0.01\bar{\sigma}$	$\{0.001, 0.01, 0.05\}\bar{\sigma}$
θ_{AUC}	AUC threshold for Stage 2 bypass	0.97	$\{0.95, 0.97, 0.99\}$

7.9 What AMBA Does That Existing Methods Do Not

Table 5: Qualitative comparison of AMBA against the existing baseline techniques.

Property	SMOTE	ADASYN	Class Wt.	EasyEns.	AMBA
Boundary-aware oversampling	×	Partial	×	×	✓
Gradient-directed synthesis	×	×	×	×	✓
Stability-aware ensemble weights	×	×	×	×	✓
Ensemble aggregation	×	×	×	✓	✓
Probability calibration	×	×	Partial	×	✓
IR-adaptive routing	×	×	×	×	✓
Adaptive non-linear f_0	×	×	×	×	✓
Scarcity-aware budget cap	×	×	×	×	✓
Diversity filter	×	×	×	×	✓
AUC-gated bypass	×	×	×	×	✓
Model-agnostic	✓	✓	Partial	✓	✓

8 Experimental Roadmap

8.1 Experimental Design

The study proceeds in five phases:

Phase 1. Data Preparation. Loading, preprocessing, and stratified splitting.

Phase 2. Baseline Evaluation. All seven models on raw imbalanced data; no correction.

Phase 3. Existing Technique Baselines. Nine baseline techniques applied to every model–dataset combination.

Phase 4. AMBA Evaluation. AMBA applied to all model–dataset combinations with nested cross-validation for hyperparameter tuning.

Phase 5. Stability and Comparative Analysis. Stability profiles, technique–model interaction heatmaps, synthetic sample distribution visualisation, and failure mode characterisation.

8.2 Phase 1: Data Preparation

Each dataset will be subjected to the following preprocessing steps:

- Missing value imputation: median for continuous features, mode for categorical
- Feature scaling via **StandardScaler** (zero mean, unit variance)
- Stratified 5-fold cross-validation split (preserves per-fold class ratio)
- Held-out test set (20 % of data) reserved for final evaluation

8.3 Phase 2: Baseline Evaluation

All seven models are trained on the raw imbalanced data with no correction applied. All metrics from Table 4 are recorded for each fold, establishing the unadjusted performance floor from which improvements are measured.

8.4 Phase 3: Existing Technique Baselines

Each of the nine existing techniques is applied to every model–dataset combination. The full result matrix comprises:

$$\underbrace{7}_{\text{models}} \times \underbrace{4}_{\text{datasets}} \times \underbrace{9}_{\text{techniques}} = 252 \text{ experiment cells.}$$

Each cell reports the mean and CV for every metric.

8.5 Phase 4: AMBA Evaluation

AMBA is applied to all model–dataset combinations with hyperparameters tuned via nested cross-validation to prevent data leakage. Each run uses $K = 10$ ensemble members and $V = 5$ stability estimation folds. The routing components are engaged automatically based on each dataset’s IR and pilot AUC.

8.6 Phase 5: Analysis

Beyond aggregate tables, Phase 5 performs four targeted analyses:

1. **Stability profiles.** Plot $\text{CV}(\text{F1})$ as a function of IR for each model family. We expect a positive correlation between IR and instability that AMBA should consistently dampen.
2. **Technique–model interaction.** Produce heatmaps of F1 improvement per technique per model to expose which pairings systematically favour or harm particular architectures.
3. **Synthetic sample distribution.** Visualise AMBA-generated samples via PCA or t -SNE to verify that boundary-directed synthesis produces samples geometrically closer to the decision boundary than standard SMOTE, and that the diversity filter prevents clustering.
4. **Routing analysis.** Characterise how often each routing path (IR-bypass, AUC-bypass, full pipeline) is triggered across model–dataset combinations, and whether the routing decisions correlate with performance outcomes.

9 Expected Contributions

This study aims to make three distinct contributions to the imbalanced learning literature:

1. **A systematic instability benchmark.** A publicly reproducible table of mean and CV metrics for 7 models $\times$ 4 datasets $\times$ 10 techniques (9 existing + AMBA). This alone fills a recognised gap in the literature.
2. **A characterisation of model–imbalance interaction.** An empirical account of how model capacity, decision-boundary geometry, and imbalance ratio jointly determine prediction stability.
3. **The AMBA framework.** A principled, model-agnostic technique that simultaneously addresses boundary coverage, prediction stability, and regime-adaptive routing, with clearly stated hyperparameters and complete pseudocode for reproducibility. AMBA is the first imbalance-handling framework to incorporate all of the following in a single coherent design: boundary-guided synthesis, gradient-directed interpolation, scarcity-proportional budget control, diversity filtering, stability-weighted ensemble aggregation, and adaptive IR/AUC routing.

10 Conclusion

This document has laid out a complete research pitch for an empirical study of model instability under minority data scarcity. We have identified four datasets spanning a wide range of imbalance ratios, selected seven diverse classifiers, defined a dual-criterion evaluation suite covering both performance and stability, catalogued existing techniques to be used as baselines, and introduced AMBA—a novel four-component framework that combines adaptive routing, boundary-guided synthetic oversampling with scarcity-proportional budget control, and stability-weighted ensemble aggregation.

The core message is straightforward: mean F1-score is an incomplete picture. A technique that halves prediction variance, even at a small cost to the mean, is often far more valuable in any deployment context where reliable minority-class detection matters. Equally important is the recognition that no single augmentation strategy is optimal across all imbalance regimes: at mild imbalance, simple threshold calibration dominates; at extreme imbalance with tiny minority counts, unconstrained synthesis causes mode collapse; when classes are already well-separated, augmentation adds noise, not signal. AMBA is designed with both principles as primary objectives—stability maximisation and regime-adaptive strategy selection.

The next step is implementation and execution of the experimental roadmap described in Section 8.

This document is a research pitch. All claims about AMBA are design-time hypotheses grounded in the motivation for each architectural decision. No experimental results are reported.